

PARAS BANSAL

Portfolio | Email: parasb736@gmail.com | Phone: +91-7889099620 | LinkedIn | GitHub

Education

Indian Institute of Information Technology, Allahabad <i>B.Tech in Information Technology; CGPA: 8.89</i>	Aug 2024 – May 2028 <i>Prayagraj, Uttar Pradesh</i>
Golden Earth Global School, Sangrur <i>Class 12 (CBSE); Percentage: 94.6%</i>	Apr 2023 – Mar 2024 <i>Sangrur, Punjab</i>
Golden Earth Global School, Sangrur <i>Class 10 (CBSE); Percentage: 97%</i>	Apr 2021 – Mar 2022 <i>Sangrur, Punjab</i>

Key Projects

Pseudo2CPP C++, NLP, Compiler Design, DSA GitHub	Feb 2026 – Mar 2026
- Implemented an NLP-driven C++ transpiler converting English algorithmic queries into optimized DSA implementations using 50+ reusable templates. - Designed a semantic keyword engine analyzing 100+ tokens to classify algorithms with 90%+ accuracy. - Built modular parsing, semantic analysis, and code generation pipelines enabling automated competitive programming boilerplate with O(1) template retrieval.	
Energy-Aware Beam-ACO Routing C++, Metaheuristics, Optimization GitHub	Oct 2025 – Dec 2025
- Implemented a hybrid Ant Colony Optimization (ACO) + Beam Search routing algorithm in C++, reducing path-planning cost by 15%. - Integrated Beam Search top-K pruning to limit exploration space, improving convergence speed by 40%. - Designed a physics-based energy cost model using velocity and drag, improving routing efficiency by 15%.	
Samadhaan MERN Stack, REST APIs, MongoDB GitHub	Jun 2025 – Aug 2025
- Developed a scalable MERN-based grievance management platform supporting 12+ civic complaint categories with real-time status tracking. - Built and documented 15+ RESTful APIs using Node.js and Express.js with JWT authentication and role-based access control. - Optimized backend performance using MongoDB indexing, reducing API latency by 20% and improving UI load time by 25%.	

Technical Skills

Languages: C++ (STL), C, Java, JavaScript (ES6+), Python (Intermediate), HTML5, CSS3, SQL
Technologies/Frameworks: Git, GitHub, Linux, React.js, Node.js, Tailwind CSS, NumPy, MongoDB, MySQL
Coursework: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Theory of Computation

Competitive Programming

CodeChef (**4-Star**, Rating 1820) | Codeforces (**Expert**, Rating 1710) | LeetCode (**Knight**, Rating 1914)

Achievements

- Global Top 5% (Expert) on Codeforces** with rating **1710**; ranked **830 / 91,904+** active users in India with best contest rank **451** in Codeforces Round 1066 (Div.1 & 2).
- Achieved **Global Rank 93** in CodeChef Starters 224 among thousands of participants worldwide.
- Secured **Global Rank 38** in LeetCode Weekly Contest 484.
- Solved **500+** problems across diverse **Data Structures & Algorithms** topics on Codeforces, CodeChef, and LeetCode.
- JEE Main 2024: **AIR 3341 (Top 0.2%** among 1.4M candidates); JEE Advanced 2024: **AIR 9232**.

Leadership & Volunteering

Head Boy Golden Earth Global School	Class 10
Led a student council of 35 members to organize 10+ school events, improving school-wide communication.	
Student Mentor Prayaas (Social Service)	2024 – Present
Mentored 100+ underprivileged students; developed resources rated 90% Excellent.	